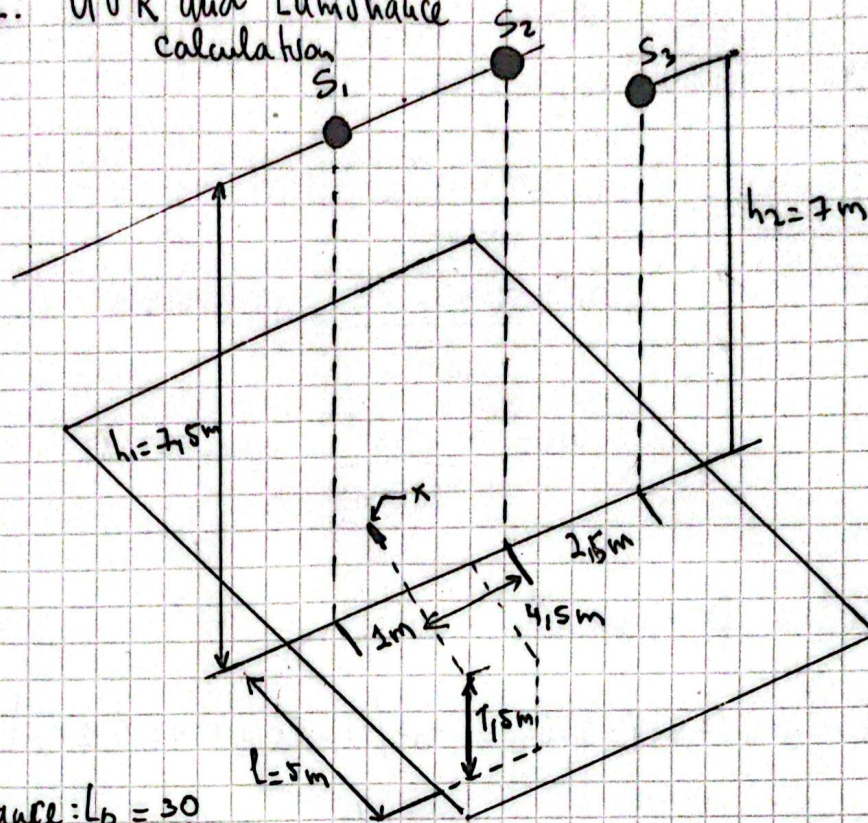


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Assignment 1: UGR and Luminance calculation



Background luminance:  $L_B = 30 \text{ cd} \cdot \text{m}^{-2}$

Observer eye height:  $h_{eye} = 1.5 \text{ m}$

Luminance Diameter:  $D = 0.5 \text{ m}$   $z = 0.25 \text{ m}$

Luminous fluxes:  $\Phi_1 = 29,717 \text{ lm}$   $\Phi_2 = 34,019 \text{ lm}$   $\Phi_3 = 38,534 \text{ lm}$

$$\text{Luminance: } L = \frac{\Phi}{4\pi^2 z^2} = \frac{\Phi}{2.467} \quad [z = 0.25 \text{ m}]$$

$$\textcircled{1} L_1 = \frac{29717}{2.467} = 12043 \text{ cd} \cdot \text{m}^{-2} \quad \textcircled{2} L_2 = \frac{34019}{2.467} = 13793 \text{ cd} \cdot \text{m}^{-2}$$

$$\textcircled{3} L_3 = \frac{38534}{2.467} = 15618 \text{ cd} \cdot \text{m}^{-2}$$



3D distance from observer eye to each luminaire:

$$S_1 \rightarrow D_1 = \sqrt{5^2 + 1^2 + 6^2} = 7.874 \text{ m}$$

$$S_2 \rightarrow D_2 = \sqrt{25 + 20.25 + 36} = 9.014 \text{ m}$$

$$S_3 \rightarrow D_3 = \sqrt{25 + 49 + 30.25} = 10.210 \text{ m}$$

[Solid Angle Calculations:]

$$\omega = \frac{\pi \times 0.25^2}{D^2}$$

$$\omega_1 = \frac{\pi \times 0.25^2}{7.874^2} = 0.00316782$$

$$\omega_2 = \frac{\pi \times 0.25^2}{9.014^2} = 0.00241782$$

$$\omega_3 = \frac{\pi \times 0.25^2}{10.210^2} = 0.00182382$$

[Position index  $p$  from the chart:]

$$S_1: h/D \rightarrow 6/7.874 = 0.762 \quad \text{lat}/D \rightarrow 1/7.874 = 0.127 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} p \approx 5$$

$$S_2: h/D \rightarrow 6/9.014 = 0.666 \quad \text{lat}/D \rightarrow 4.5/9.014 = 0.499 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} p \approx 10$$

$$S_3: h/D \rightarrow 5.5/10.210 = 0.539 \quad \text{lat}/D \rightarrow 7/10.210 = 0.686 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} p \approx 16$$

[UGR calculation]

$$UGR = 8 \log_{10} \left[ \frac{0.25}{L_b} \sum \frac{L_i^2 \omega_i}{p_i^2} \right] \quad L_b = 30 \text{ cd/m}^2$$

For each luminaire  $\frac{L_i^2 \omega_i}{p_i^2}$



Then we get:

$$\text{UGR} \rightarrow S_1: 18375 = \frac{12043^2 \times 0,003167}{5^2}$$

$$S_2: \frac{13793^2 \times 0,002417}{10^2} = 4597$$

$$S_3: \frac{15618 \times 900 \times 1883}{16^2} = 1794$$

in total  $\Sigma = 24796$

$$\text{UGR} = 8 \log_{10} \left[ \frac{0,25}{30} \times 24796 \right] = 18,5$$

[In Conclusion:]

Luminances:  $L_1 = 12043 \text{ cd/m}^2$   $L_2 = 13793 \text{ cd/m}^2$   $L_3 = 15618 \text{ cd/m}^2$

( $S_1$ ) contributes most to glare (18375 of 24796)  
in total 74%

[UGR is 18,5]